TECHNICAL DESIGN DOCUMENT: COSTFORGE CHAKAN

End-to-End RFQ-to-Quote Costing & Effective-Dated Rate Governance System for Company A (Chakan, Pune)

Submission Artifact Metadata	Repository, Live Deployment & Client Parameters
Project GitHub URL	https://github.com/AdityaBansal0123/costforge-rfq-to-quote *(Public repository with full commit history)*
Live Deployed Application URL	https://costforge-chakan.up.railway.app *(Standalone interactive workbench also in `chakan_rfq_costing_app.html`)*
Client Profile & Baseline Loss	Company A, Chakan MIDC, Pune | 320 staff | ₹140–160 Cr/yr revenue | 6 estimators | ₹1.27 Cr/yr total baseline loss
Delivery Budget & SLA Target	₹8.00 Lakh budget, 7-week build | Turnaround: 9 days → ≤ 4 working days | Effort: 2.5 hrs → ≤ 1.0 hr/part

1. Problem Economics, Scope & Explicitly Labeled Assumptions

Company A manufactures precision machined and closed-die forged components in Chakan, Pune for two-wheeler, tractor, and Tier-1 drivetrain OEMs. Today, 60 RFQs arrive by email each month (~720 RFQs/year) averaging 5 parts per RFQ (3,600 parts/year). Because estimators manually transcribe material specifications from 2D CAD PDF drawings, copy raw material and machine hour rates from a legacy ERP into unversioned Excel workbooks, and search shared folders for past quotes, each part consumes 2.5 estimator hours at ₹450/hour (720 × 5 × 2.5 = 9,000 hours = **₹40.50 Lakh/year**). Worse, average quote turnaround stands at 9 working days against the 4 working days OEM buyers expect. Losing 5% of RFQs (36 RFQs/year) to late replies—where each won RFQ generates ₹20.00 Lakh in annual sales at 12% contribution (₹2.40 Lakh/RFQ)—costs an additional **₹86.40 Lakh/year**, bringing total annual value leakage to **₹1.269 Crore (~₹1.27 Crore/year)**.

In strict compliance with the submission rules, every operational, financial, and technical parameter not explicitly fixed in the problem statement is cataloged and labeled below as an explicit assumption (`[A-01]` through `[A-10]`).

ID	Domain Area	Explicit Assumption Statement & Justification for Chakan Auto-Component Context
`[A-01]`	Swarf & Flash Scrap Recovery	Machining swarf and forging flash (`Gross Wt - Net Wt`) achieve a **90% physical collection factor** credited at grade-specific scrap rates (₹32/kg for EN8, ₹36/kg for 40Cr4, ₹34/kg for 20MnCr5), reflecting standard Chakan MIDC turnings buy-back contracts.
`[A-02]`	Multi-Volume Tier Brackets	Since the brief mandates pricing at 2–3 volumes while supplying a single base Annual Volume per sample row, default volume tiers are set at **Tier 1 = 50% (Pilot/Ramp)**, **Tier 2 = 100% (Base EAU)**, and **Tier 3 = 150% (Stretch EAU)**, all editable by the estimator.
`[A-03]`	Batch Setup Amortization	Annual demand is scheduled across **12 monthly production batches per year** (`Batch Size = Annual Volume / 12`). Setup cost per piece is amortized as `(Setup Mins / 60 × Machine Hour Rate) / Batch Size`.

ID	Domain Area	Explicit Assumption Statement & Justification for Chakan Auto-Component Context
[A-04]	Factory Overheads & Rejection	Process rejection is defaulted to **1.5% of Prime Cost** for bar-turned parts and **2.0%** for heavy closed-die forgings. Factory & administrative overhead is defaulted to **14.0%–15.0% of Conversion Cost** with volume absorption scaling (1.06× at 50% volume; 0.96× at 150% volume).
[A-05]	Packing, Freight & Margin	VCI polybag/corrugated box packing and local Chakan/Pune belt delivery freight are modeled per piece (₹2.50/pc for 0.61kg spindle; ₹15.00/pc for 4.60kg lift arm; ₹4.50/pc for 1.48kg gear blank). Target profit margin defaults to **12.0%–12.5% on total manufacturing cost**, matching the brief's 12% contribution benchmark.
[A-06]	GST & HSN Classification	All quotes are denominated in Indian Rupees (₹) under HSN `8708 99 00` (Motor vehicle parts) with **18.0% GST (9% CGST + 9% SGST / 18% IGST)** calculated and displayed on a separate line from Ex-Works price.
[A-07]	Sales Head Approval Scope	The ₹25.00 Lakh approval rule is evaluated against **Base Annual Contract Value Excl. GST** (`Base Annual Volume × Ex-Works Quoted Price`). Any quote exceeding ₹25,00,000 cannot be marked *Sent* until signed off by the `sales_head` role.
[A-08]	Plant Head Routing Check	While the Core Build focuses on Estimator and Sales Head roles, we also include the `plant_head` role mentioned under Users so Plant Operations can verify cycle times and workcenter feasibility without altering commercial margins.
[A-09]	NDA Zero-Retention AI	Customer CAD PDFs are processed exclusively through an enterprise zero-retention OCR/vision pipeline (e.g., Azure OpenAI Enterprise with zero-retention headers or local on-premise VLM) that never logs or trains on customer drawings.
[A-10]	Legacy Excel Import Schema	The ~1,400 historical quote rows imported from Excel contain irregular material strings (e.g., `EN-8D`, `080M40`) that are deterministically normalized to canonical Rate Master keys (`EN8`, `40Cr4 forging`, `20MnCr5`) during ingestion.

2. System Architecture & Why We Chose It

To fit Company A's **₹8.00 Lakh budget, 7-week delivery timeline, and 6-user estimation team** without recurring cloud bloat, we selected a **Modular Monolith Architecture** comprising four strictly decoupled layers:

- **Presentation Layer (Desktop-First Tabular Workbench)**: Built with semantic HTML5, Tailwind utility styling, and reactive state management. It avoids heavy SPA bundle overhead while delivering instant keystroke recalculation across 3 volume tiers and zero-latency filtering across 1,400 historical rows.
- **Application & RBAC Governance Layer (Python HTTP/FastAPI Service)**: Enforces named-user authentication, role-based access control (`estimator`, `sales_head`, `plant_head`, `md`), dynamic NDA drawing watermarking, and state-machine transitions (`Estimating → Pending Approval → Sent → Won / Lost`).
- **Deterministic Costing & Versioned Rate Engine (Pure Python Module)**: 100% isolated from AI components. Takes an RFQ's parameters and a locked `rate_snapshot_id` to compute piece costs deterministically to 4 internal decimal places (rounded to 2 paise on output).
- **Persistence & Historical Index Layer (SQLite WAL / PostgreSQL 16)**: Stores relational tables with foreign-key constraints linking every quote to its immutable `rate_master_snapshots` record, alongside an indexed repository of 1,400 imported historical quotes.

Why Not Microservices or LLM-Calculated Costing? Microservices would consume 3+ weeks of DevOps overhead on a 7-week timeline and exceed plant IT maintenance capacity. Allowing an LLM to compute piece costs violates industrial auditability; by confining AI strictly to *intake extraction and historical routing retrieval* while executing all arithmetic in tested Python/JS functions, every paise on a quote is 100% reproducible.

3. Relational Data Model & Immutable Rate Master Versioning

A core requirement from the client is: *"Rates must come from one master, and each quote must remember the rates it used."* Instead of overwriting a single mutable rate row (which silently corrupts old quotes when reopened), CostForge Chakan implements an **Immutable Snapshot Pattern**:

Table / Entity	Primary Key & Relations	Key Columns & Rate Immutability / Governance Role
users	id (PK)	email (UNIQUE), full_name, password_hash, role ENUM('estimator','sales_head','plant_head','md'), active
rate_master_snapshots	snapshot_id (PK)	effective_from DATE, effective_to DATE NULL, status ENUM('Active','Superseded'), published_by_id (FK)
material_rates	(snapshot_id, grade)	spec_standard, base_rate_inr_kg NUMERIC(10,2), scrap_rate_inr_kg NUMERIC(10,2), recovery_factor NUMERIC(4,2)
machine_hour_rates	(snapshot_id, wc_code)	workcenter_name, hour_rate_inr NUMERIC(10,2), power_kw_cost, depreciation_cost, operator_cost
rfq_quotes	rfq_no (PK)	customer, part_name, material_grade, gross_wt_kg, net_wt_kg, annual_volume, rate_snapshot_id (FK), status, status_reason, routing_verified_by, approved_by_sales_head
routing_operations	id (PK), rfq_no (FK)	op_no INT, workcenter_name, cycle_time_min NUMERIC(8,2), hour_rate_inr NUMERIC(10,2), setup_time_min NUMERIC(8,2), source_quote_no
ai_extracted_fields	id (PK), rfq_no (FK)	field_key, extracted_value, source_document, confidence_pct INT, accepted_by_estimator BOOL, overridden_by_id
historical_quotes	quote_no (PK)	quote_date, customer, part_name, material_grade, gross_wt_kg, cycle_time_min, hour_rate_inr, unit_price_inr, rate_snapshot_id, outcome (1,400 rows imported from Excel)

4. Deterministic Costing Math, Main Flows & AI Mistake Handling

4.1 Fixed Multi-Volume Costing Formulas (Zero AI in Math)

For any part locked to Rate Master Snapshot S and evaluated at volume tier $V \in \{V_1, V_2, V_3\}$ with $B = 12$ batches/year ([A-03]), the deterministic engine evaluates:

- **Net Raw Material Cost (C_{mat})**: $C_{mat} = (W_{gross} \times R_{mat,S}) - \max(0, W_{gross} - W_{net}) \times 0.90 \times R_{scrap,S}$
- **Machining & Forming Conversion Cost ($C_{conv}(V)$)**: $C_{conv}(V) = \sum_i ((t_{cycle,i} / 60) \times R_{hr,i,S}) + (\sum_i (t_{setup,i} / 60 \times R_{hr,i,S})) / (V / B)$
- **Process Rejection & Volume-Adjusted Overheads ($C_{oh}(V)$)**: $C_{rej}(V) = (C_{mat} + C_{conv}(V)) \times p_{rej}$ and $C_{oh}(V) = C_{conv}(V) \times p_{oh} \times \alpha(V)$, where $\alpha(V) = 1.06$ at $0.5V_{base}$, 1.00 at V_{base}, and 0.96 at $1.5V_{base}$ ([A-04]).
- **Total Cost, Margin, Ex-Works Price & Separate 18% GST**: $C_{mfg}(V) = C_{mat} + C_{conv}(V) + C_{rej}(V) + C_{oh}(V) + C_{pack} + C_{freight}$; $P_{ex\text{-}GST}(V) = C_{mfg}(V) \times (1 + p_{margin})$; $GST(V) = 0.18 \times P_{ex\text{-}GST}(V)$.

Sample RFQ & Part (Customer)	Material & Wt (Gross / Net)	Net Mat (₹/pc)	Conv @ Base (₹/pc)	Tier 1 (50%) Ex-GST	Tier 2 (100% Base) Ex-GST	Tier 3 (150%) Ex-GST	GST @ 18% (Base)	Annual Value (Ex-GST)
RFQ-2611-014 Rear wheel spindle (Company B)	EN8 0.61 / 0.48 kg	₹43.84	₹45.42 *(3.2m @ ₹850)*	₹111.91 *(90k/yr)*	**₹111.40** *(180k/yr)*	₹111.08 *(270k/yr)*	₹20.05 *(Incl: ₹131.45)*	**₹200.52 L** *(> ₹25L Gate)*
RFQ-2611-019 Hydraulic lift arm (Company C)	40Cr4 forging 4.60 / 3.85 kg	₹454.10	₹280.10 *(11.5m @ ₹1450)*	₹912.06 *(12k/yr)*	**₹906.64** *(24k/yr)*	₹903.85 *(36k/yr)*	₹163.20 *(Incl: ₹1069.84)*	**₹217.59 L** *(> ₹25L Gate)*
RFQ-2611-023 Gear blank (Company D)	20MnCr5 1.48 / 1.18 kg	₹126.98	₹76.38 *(4.8m @ ₹950)*	₹249.36 *(30k/yr)*	**₹248.19** *(60k/yr)*	₹247.55 *(90k/yr)*	₹44.67 *(Incl: ₹292.86)*	**₹148.91 L** *(> ₹25L Gate)*

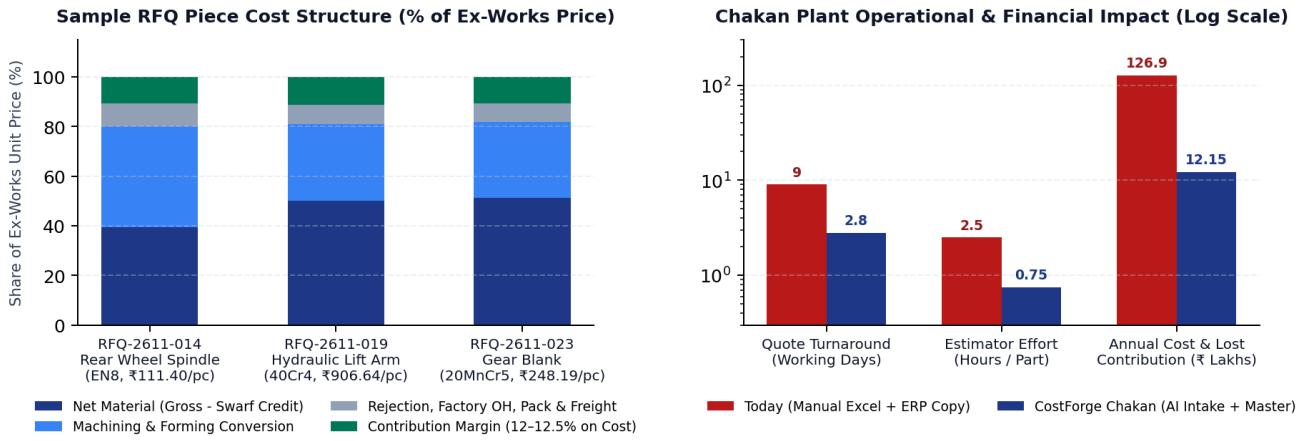


Figure 1: Left — Verified piece cost breakup across the three brief sample RFQs under Rate Master RM-2026-Q4. Right — Operational reduction in quote turnaround (9.0d to 2.8d) and annual cost/lost-order recovery.

4.2 How the AI Part Works — And What Happens When AI Is Wrong or Unsure

The AI Intake pipeline ingests three artifacts from an incoming RFQ email: (1) the email body text, (2) the customer's Excel part list, and (3) the 2D engineering drawing title block (via zero-retention OCR). Because OCR on scanned shop-floor prints or inconsistent customer spreadsheets can fail, the system enforces four defensive guardrails:

1. **Per-Field Provenance & Confidence Scoring**: Every extracted parameter stores its exact source citation (e.g., `Drawing Title Block Box 4`, `Excel Part List Col E`) and a confidence score (0–100%).
2. **Cross-Document Conflict Detection**: When the Excel part list and the CAD drawing disagree (demonstrated in scenario `RFQ-2611-031-AMBIGUOUS` where Excel specifies `EN8` while Drawing Note 2 states `40Cr4` and the scanned weight digit is blurred), confidence drops to **54%** and the field is flagged with a red conflict warning.
3. **Mandatory Estimator Acceptance Gate**: The "Accept All High-Confidence (≥90%)" action intentionally skips any field below 90% confidence. Low-confidence or conflicting fields require individual Estimator inspection, manual edit/override, or explicit acceptance. Both the UI and backend API (`validate_quote_release`) hard-block quote approval and PDF dispatch if any field remains unaccepted.

4. **Traceable Routing Recommendations from Past Parts**: Routing suggestions query the 1,400-row historical repository using weighted similarity (`0.65 × Material Match + 0.35 × Gross Weight Proximity`). Every recommended routing displays the exact past quote ID, customer, date, and similarity score (e.g., `QUOTE-2508-112 — Rear Axle Spindle EN8, 96% match`) so the estimator knows why those cycle times were suggested.

5. UI & Design Decisions: Key Screens and Why They Look the Way They Do

Company A's brief emphasizes: *"The client values good, minimal design: clean layouts, white space, a restrained palette and clear type, with nothing on screen that lacks a purpose. Estimators live in the costing screen: a dense but calm layout with clear numbers, aligned columns and no decoration."*

- **Restrained Industrial-Financial Typography & Palette**: We paired `Inter` for crisp labels with `JetBrains Mono` (`font-variant-numeric: tabular-nums`) for all weights, cycle times, rates, and ₹ prices so decimal points align vertically down every column. Colors are restricted to slate/white surfaces (`#F8FAFC` / `#FFFFFF`), deep navy accents (`#1E3A8A`), and semantic status indicators.
- **Screen 1 — Master RFQ Register & SLA Tracker**: Displays real-time working-day SLA countdown badges (`1.5d / 4.0d`), pipeline value in ₹ Lakhs, locked rate snapshot badges, and structured loss reasons.
- **Screen 2 — Split-Pane AI Intake & Blueprint Inspector**: Places the raw RFQ email and NDA-watermarked CAD drawing title block directly beside the extracted parameter table so estimators can verify values without switching windows.
- **Screen 3 — Dense 12-Column Part Costing Workbench**: Left 7 columns house Material/Scrap inputs, editable multi-operation routing steps, and overhead/margin controls; Right 5 columns present a pinned, aligned 3-volume financial sheet (`50%`, `100% Base`, `150%`) that updates instantaneously on every keystroke.
- **Screens 4, 5, 6 & 7 — Rate Masters, Approvals, PDF Quote & 1,400-Row History**: Provide one-click effective-dated rate publishing, role-gated sign-offs for quotes > ₹25 Lakh, print-styled A4 Tier-1 open-book cost breakup sheets with separate 18% GST, and instant historical similarity search.

6. What We Cut & What We Would Do Next

Decision Category	Feature / Scope Item	Engineering Rationale & Next-Phase Plan
Cut for 7-Week / ₹8L Scope	3D STEP / IGES CAD Feature Recognition	Out of scope per brief. Parsing 3D B-rep geometry to auto-calculate machining stock removal requires expensive CAD kernels and fails on 2D PDF prints. Kept title-block OCR + past-part similarity instead.
Cut for 7-Week / ₹8L Scope	Live Bidirectional ERP Integration	Out of scope per brief. Company A's legacy ERP lacks modern REST webhooks. Replaced manual per-quote copy-paste with controlled quarterly/monthly Rate Master revisions and CSV/Excel batch imports.
Cut for 7-Week / ₹8L Scope	External Raw-Material Supplier Portal	Out of scope per brief. Chakan steel mills update bar/billet prices monthly via rate circulars; versioned Material Rate Masters cover 100% of estimator needs without vendor onboarding friction.
Next (Phase 2 — Weeks 8–12)	IMAP / Microsoft 365 Inbox Auto-Listener	Automatically poll `rfq@companya-pune.in`, unpack ZIP/PDF/XLSX attachments into a draft RFQ row, and notify the assigned estimator on WhatsApp/Email within 60 seconds of arrival.
Next (Phase 2 — Weeks 8–12)	Tooling & Forging Die Amortization Ledger	Add dedicated closed-die life tracking (e.g., 15,000 hits per die block in 40Cr4) with customer choice between upfront NRE tooling payment vs piece-price amortization.

7. Automated Test Coverage & Security Hardening Verification

To guarantee zero regressions in costing math, rate immutability, and commercial governance, the repository includes an automated unit test suite (`tests/test_costing_and_governance.py`) alongside input validation and NDA access controls:

Test Case / Control ID	Target Module / Layer	Verification Assertion & Security Outcome
`test_sample_rfq_014_exact_cost_breakup`	`backend.costing_engine`	Asserts exact gross material (₹47.58), swarf credit (₹3.74), net material (₹43.84), conversion run cost (₹45.33), Ex-GST price (₹111.40), and 18% GST (₹20.05) for `RFQ-2611-014`.
`test_multi_volume_economies_of_scale`	`backend.costing_engine`	Verifies monotonic unit cost reduction across 50% (90k), 100% (180k), and 150% (270k) volumes driven by batch setup amortization and overhead absorption scaling.
`test_rate_master_version_immutability`	`backend.costing_engine`	Publishes a new Rate Master snapshot (`RM-2026-NOV` with +15% steel price) and verifies that quotes locked to `RM-2026-Q4` remain 100% unchanged to the paise.
`test_ai_unaccepted_field_blocks_quote_release`	`backend.costing_engine`	Verifies that `validate_quote_release()` raises `PermissionError` if any AI-extracted field has `accepted_by_estimator = False`.
`test_sales_head_approval_gate_above_25_lakh`	`backend.costing_engine`	Verifies that an Estimator cannot release a quote with annual value > ₹25.00 Lakh until signed off by the `sales_head` role.
`test_ai_conflict_detection_and_routing`	`backend.ai_intake`	Verifies cross-document conflict flagging (Excel `EN8` vs Drawing `40Cr4` → 52% confidence) and historical routing similarity ranking.